Assignment module – 2

1. Overview of c programming

the c programing language is one of the most important and influential programing language.

It was developed in the early 1970’s by dennis Ritchie at bell labs.

C programing is important because it forms the functions for many famous programming languages like python, java, c++ .

Many concept like loops functions variable, and pointer used in modern language come directly from c.

The first major used of c was in rewriting the unix os since unix was written in c it became easier to move the os from one computer system to another compute system.

Another reason c remains popular is its educational value learning c helps student how computer works internally including memory management and data handling.

In conclusion c remains one of the most powerful and widely used programming language in the world…

Lab ex

1. Embedded system
   - C is used in device like washing machine, cars, robots, and smart tv because it is feel and efficient.
2. Operating system
   - Os link linux and windows use c for system level programming.
3. Game development
   - C is used in game and graphics programming for high performance.

2. Setting up environment

1. Install gcc compiler
2. Add gcc bin path to environment variable
3. Check installation using

Ide options

Dev c++ - install and start coding

Codblocks - install codeBlocks with minGW setup

Visual studio code - install vs code add c extenstion and concept gcc compiler

Conclusion :-

Compiler converts c code into executable programs, and ides make coding easier

3.basic structure of c programming

A c program consists of different parts such as header files main function comments data types and variable.

Header file :- used to include in library functions examples include<stdio.h>

Main functions :- execution of program start from main()

Comments:- used to explain code

Data types:- defines the types of data like int float char

Variables:- used to store values in memory

Printf():- function display output on screen

Return 0:-indicates successful completion of program \

4. Operator in c

- Arithmetic
- Relational
- Logical
- Assignment

- Increment/Decrement
- Bitwise
- Conditional (Ternary)
- Special Operators

1. arithmetic

Operator	**Meaning**	**Example**
+	Addition	a + b
-	Subtraction	a - b
*	Multiplication	a * b
/	Division	a / b
%	Modulus (remainder)	a % b

2. Relational

Operator	**Meaning**
==	Equal to
!=	Not equal to
>	Greater than
<	Less than
>=	Greater than or equal
<=	Less than or equal

3. Logical

Operator	**Meaning**
&&	Logical AND
`	
!	Logical NOT

4. Assignment

Operator	**Example**
`=`	`a = 5`
`+=`	`a += 2`
`-=`	`a -= 2`
`*=`	`a *= 2`
`/=`	`a /= 2`

5. Increment/Decrement

Operator	**Meaning**
`++`	Increment
`--`	Decrement

6. Bitwise

Operator	**Meaning**
&	Bitwise AND
`	`
^	XOR
~	NOT
<<	Left shift
>>	Right shift

7. Conditional (Ternary

condition ? expression1 : expression2;

8. Special Operators

- 🎬 sizeof

- & (address operator)
- * (pointer operator)

5. Control flow statement in c

i] explain decision-making satements in C (if,else,nestedif-else,switch).provide example of each.

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1] if statement:-

à the if statement executes a block of code only when a condition is true.

2] if-else statement:-

à the if-else statement executes a block of code only when a condition is true and another block if it is flase.

3] nested if-else statement:-

à a nested if-else means using one if or else statement inside another.

4] switch statement:-

à the switch statement used to select one block of code form many options.

6. looping in C

i] compare and contrast while loops,for loops,and do-while loop.

1. while loop:-

à a while loop as long a condition is ture.

2. for loop:-

à a for loop is typically used when the number of repetition is known.

3.do-while:-

à a do while loop executes the code at least once,then checks the condition.

7. Loop control statement

1. break Statement

- Terminates the loop immediately.
- Control jumps to the statement after the loop.

2. continue Statement

- Skips the current iteration.
- Moves control to the next iteration of the loop.

3.. gotoStatement (rarely used)

- Transfers control to a labeled statement.
- Not recommended for regular loop control (can make code confusi

8. Functions in c

- Library functions
- User defined function

1. library function

Function	**Purpose**
Printf()	Print output to screen
scanf()	Take input from user

Function	Purpose
getchar()	Read one character
putchar()	Print one character

3. User defined function

Part	Meaning
Function Declaration	Tells compiler about function
Function Definition	Actual code of function
Function Call	Executes the function